1. Plot the maximum vertical displacement, ymax, of the beam as a function of time. Depending on your coordinate system, ymax may be negative. Does ymax eventually reach a steady value? Examine the accuracy of your simulation against the theoretical prediction from Euler beam theory:

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Theoretical maximum deflection: 0.038045 m
Simulated maximum deflection: -0.037109 m
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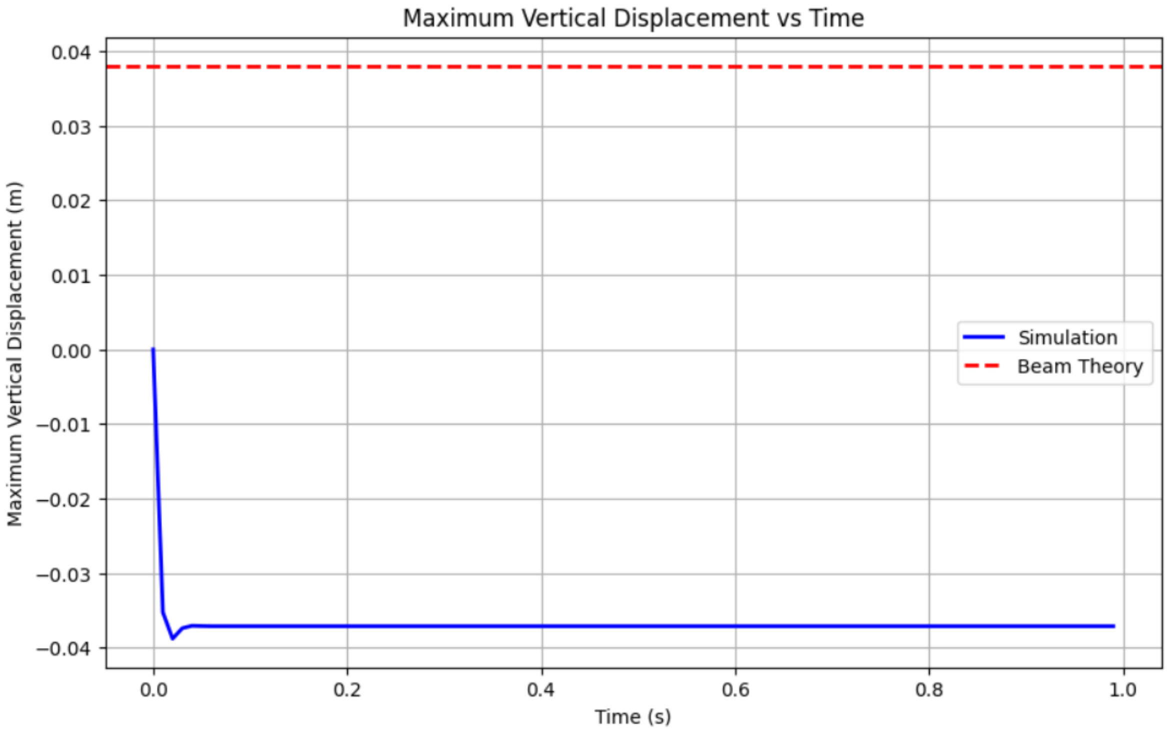
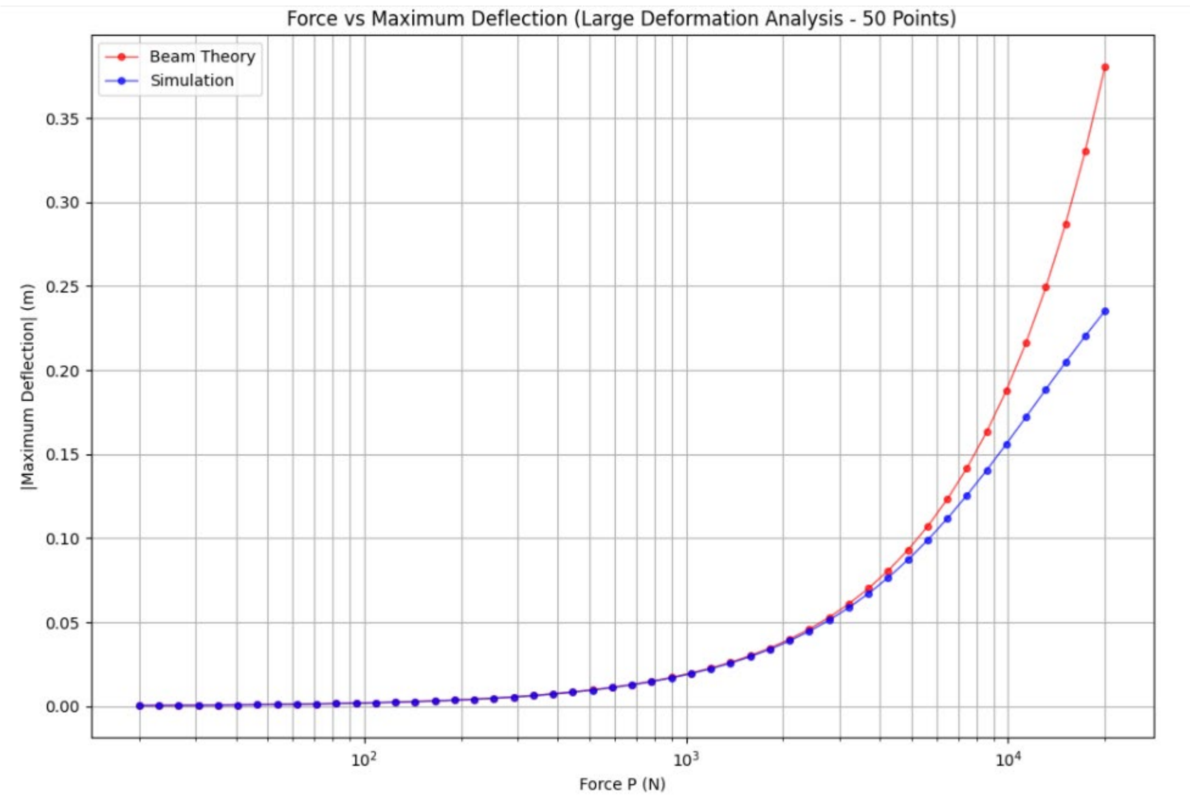


The requested plot is shown above. As stated in the problem, Ymax appears with a negative sign due to the chosen coordinate system. Comparing the magnitudes of the simulation and theoretical results, the simulation reaches steady state in less than 0.05 seconds, with a final displacement magnitude of 0.037109 m. The theoretical value obtained from the analytical formula is 0.038045 m, corresponding to a 2.5% error. This close agreement demonstrates the accuracy of both the analytical model and the simulation for small-scale deformations.

2. What is the benefit of your simulation over the predictions from beam theory? To address this, consider a higher load P = 20, 000N such that the beam undergoes large deformation. Compare the simulated result against the prediction from beam theory in Eq. 5. Euler beam theory is only valid for small deformation whereas your simulation, if done correctly, should be able to handle large deformation. You should create a plot of P (20 N ≤ P ≤ 20, 000 N) vs. ymax using data from both simulation and beam theory and quantify the value of P where the two solutions begin to diverge.?

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First significant divergence point found at P ≈ 7455 N
Relative error: 11.4%
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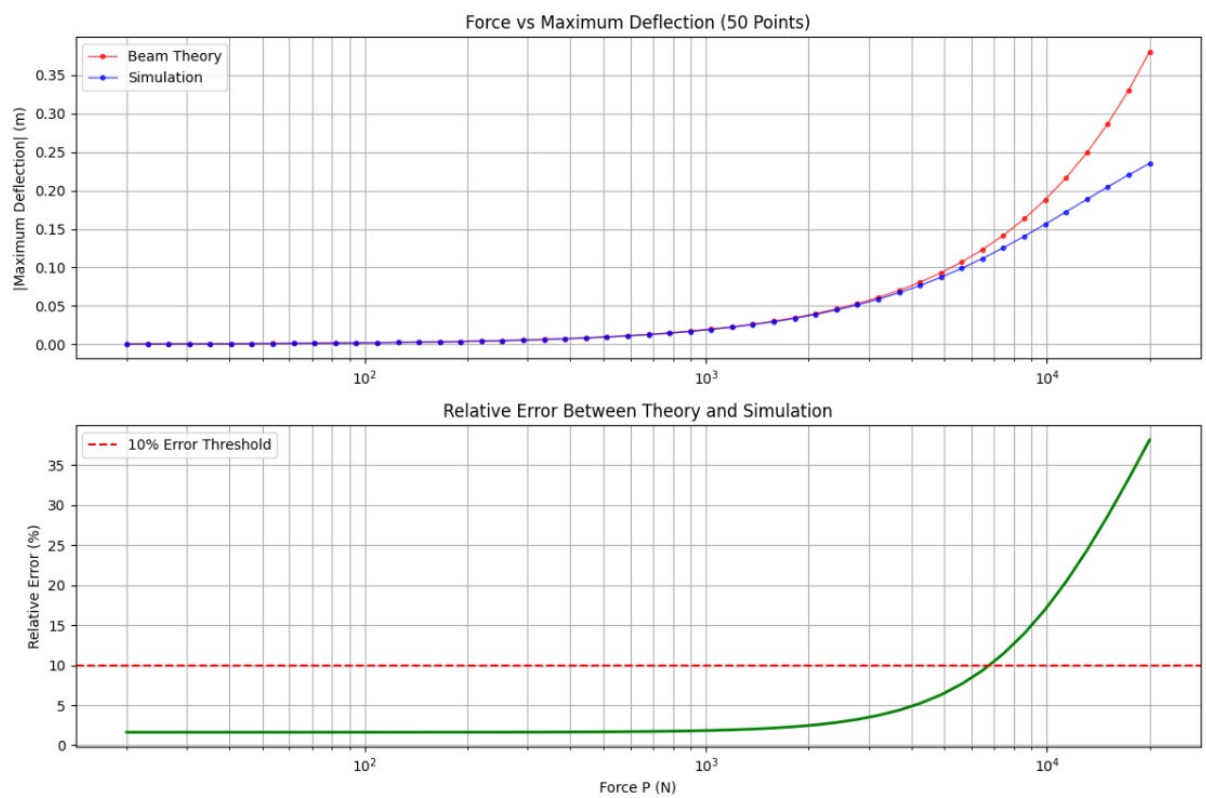

The simulation offers a significant advantage over Euler–Bernoulli beam theory by accurately modeling large-deformation effects that the analytical approach cannot capture. As seen in the plot, both methods agree well at small loads, but they begin to diverge noticeably around $P \approx 7455$ N, where the relative error reaches about 11.4%. Beyond this point, the beam theory increasingly underestimates the deflection because it assumes small rotations and linear strain–displacement relations. At higher loads, such as $P=20{,}000$N, the simulation continues to provide reliable results by incorporating geometric nonlinearities, making it better suited for analyzing large deflections where classical beam theory is no longer valid.